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Signature Verification System Using Object Detection and Machine Learning

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***Abstract*—Signature verification is a critical task in authentication systems, ensuring the security and reliability of financial transactions, legal documents, and identity verification processes. Traditional methods often struggle with variations in handwriting, forgery attempts, and the scalability required in real-world applications. Existing approaches show significant gaps, particularly in achieving high accuracy across diverse signature styles and effectively distinguishing between genuine and forged signatures. This paper highlights the necessity of a robust and intelligent system that leverages object detection techniques in conjunction with machine learning models to enhance verification accuracy. By utilizing feature extraction, deep learning architectures, and pattern recognition, the proposed system can effectively capture the unique spatial and structural characteristics of handwritten signatures. Furthermore, the integration of user-friendly interfaces and real-time verification modules ensures adaptability for practical deployment in banking, corporate, and governmental sectors. This research aims to contribute toward developing an efficient, reliable, and secure signature verification system capable of mitigating forgery and enhancing trust in authentication processes.**

***Index Terms*—Image Recognition, Object Detection, Machine Learning, Verification System**

I. Introduction

Signature verification remains one of the most essential and intricate tasks in the domain of security and authentication, particularly in financial, legal, and administrative processes. For decades, handwritten signatures have served as a widely accepted means of validating identity, and they continue to attract the attention of researchers, security experts, and in- stitutions due to their practical significance and vulnerability to forgery [1]. The act of verification encapsulates a diverse range of challenges, impacting areas such as consistency in writing patterns, variability in execution, and the capacity to distinguish genuine samples from skilled counterfeits. As we delve into the complexities of signature verification, we uncover not only the technical difficulties it presents but also the unique individuality reflected in each person's signature, making it both a personal identifier and a subject of rigorous computational study.

This study aims to provide a comprehensive examination of signature verification, highlighting the multiple dimensions that shape its effectiveness in real-world scenarios. By exploring the latest advancements in object detection, feature extraction, and machine learning techniques [2], we aspire to contribute toward developing systems with higher accuracy, adaptability, and reliability. The increasing occurrence of signature-related fraud has amplified the demand for automated solutions that can improve detection rates while reducing human error. Our approach not only focuses on the algorithmic and biometric aspects but also examines how these methods interact with broader technological and societal requirements, creating a holistic framework for robust authentication. Additionally, the discussion surrounding signa- ture verification goes beyond technical applications, involving industries and organizations dedicated to enhancing digital trust, security, and fraud prevention. Through the integration of research findings, experimental studies, and practical consid- erations, this paper seeks to promote a deeper understanding of the realities associated with signature-based authentication. In doing so, we aim to contribute toward building more secure and trustworthy verification systems that safeguard individuals and institutions alike. In the sections that follow, we will navigate through the technical and practical landscape of signature verification, guided by the overarching goal of improving accuracy, reliability, and resilience against forgery attempts.

II. MACHINE LEARNING AND SIGNATURE VERIFICATION

Machine Learning (ML) has become a powerful approach in advancing signature verification, an essential component of identity authentication in financial, legal, and organizational applications [5]. Handwritten signatures pose significant chal- lenges due to natural variation, environmental factors, and attempts at forgery. ML techniques provide reliable solutions by learning from signature data and improving verification accuracy. Below are several ways in which machine learning methods contribute to signature verification:

A. Early Identification and Screening

ML algorithms can classify signatures as genuine or forged by analyzing stroke patterns, writing flow, and geometric structures. These models detect inconsistencies that may not be visible to the human eye, allowing early detection of fraudulent signatures [**?**].

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B.Behavioral Feature Analysis

In addition to visual similarity, ML models can examine behavioral traits such as signing speed, stroke sequence, and pen pressure. These features capture the natural dynamics of a signer's handwriting, making them difficult for forgers to reproduce consistently [6].

Structural Pattern Recognition

Object detection and ML techniques can be applied to extract structural features of signatures, including curvature, spacing, alignment, and proportionality. These elements form a unique biometric signature profile that aids in distinguishing genuine signatures from imitations [4].

C.Style and Consistency Evaluation

Machine learning models can study stylistic variations across multiple samples from the same individual. Consistency in letter formation, slant, and size provides strong indicators of authenticity, while irregularities may signal forgery [7].

D.Noise Reduction and Preprocessing

Since signatures can be affected by differences in pen type, paper quality, or scanning resolution, ML-based preprocessing techniques such as normalization and feature scaling help reduce noise. This ensures that classification models perform more accurately under varying conditions [5].

III.LITERATURE REVIEW

Research in signature verification has evolved significantly over the years, moving from handcrafted feature extraction techniques to modern machine learning and deep learning approaches. This section reviews key studies from recent years, emphasizing their methodologies, datasets, performance outcomes, and limitations.

A.Classical Approaches

Earlier studies (2018–2020) relied on handcrafted texture descriptors such as Local Binary Patterns (LBP) and Uniform Local Binary Patterns (ULBP). In a 2019 study, researchers applied these descriptors with Nearest-Neighbor classifiers on Bangla and Hindi signature datasets, reporting an overall accuracy of 67.28% (Bangla 65.88–66.18%, Hindi 75.53%) [1]. While effective in capturing local structural information, these methods performed poorly against skilled forgeries and lacked scalability. Such findings motivated the shift toward automated, learning-based methods.

B.CNN-Based Models

With the rise of deep learning around 2020, Convolutional Neural Networks (CNNs) began to dominate signature verification research. A study conducted in 2020 demonstrated that CNNs trained with augmented signature samples achieved up to 97.39% accuracy, with precision (96.79%), recall (97.00%), and F1-score (97.20%) [5]. These results highlighted the effectiveness of end-to-end feature extraction and classification. However, the study noted limitations due to dataset homogeneity, raising concerns about cross-dataset generalization and overfitting.

C.Object Detection and Hybrid Methods

Between 2021 and 2023, hybrid approaches combining object detection and deep learning emerged. In 2021, researchers integrated YOLOv5 with lightweight CNNs such as MobileNet and ensembles of ResNet50, InceptionV3, and VGG19. Their work reported accuracies of 89.8% on Kaggle, 85.4% on CEDAR, 86.2% on ICDAR, and 87.5% on SigComp datasets [8], [12]. Similarly, a 2022 study demonstrated the benefits of systematic preprocessing steps such as resizing, normalization, and augmentation in improving cross-dataset robustness. Despite these achievements, such models demanded longer training times and high computational resources.

D.Insights from Handwritten Character Recognition

Research in handwritten character recognition has also contributed valuable insights. In 2020, a study on Hindi character recognition using CNNs and transfer learning with VGG16 achieved 91.25% detection accuracy and 89.9% recognition accuracy. In 2021, another study extended this work to Telugu and Hindi characters, reporting VGG16 accuracy of 93.1% for Telugu [4], [6]. These findings suggest that detection-first pipelines and transfer learning are beneficial when dealing with variable handwriting styles. The parallels with signature verification are clear, as preprocessing and segmentation play an equally vital role in improving system reliability.

E.Challenges and Limitations

Despite progress, several challenges persist. Studies published between 2020 and 2023 consistently report limited generalization across datasets. Models often achieve high accuracy on a specific dataset but degrade when tested on signatures collected under different conditions. Dataset scarcity is another concern, as most public datasets lack diversity in terms of scripts, cultural variations, and skilled forgeries. Computational complexity also remains a barrier: while ensemble-based models from 2021–2022 achieved strong accuracy, their heavy training costs make them less suitable for real-time deployment.

F.Synthesis and Future Directions

The literature from 2019–2023 highlights a clear trajectory: from handcrafted feature descriptors toward

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CNN-based sys- tems and, more recently, hybrid object detection-classification frameworks. Current research emphasizes the need for cross- dataset evaluation, domain adaptation, and lightweight archi- tectures for mobile or edge deployment. Future studies should focus on curating larger multilingual datasets, incorporating advanced forgery attempts, and developing adaptive models that learn continuously from new data. The integration of object detection with machine learning remains a promising approach for advancing accuracy, scalability, and practical deployment of signature verification systems

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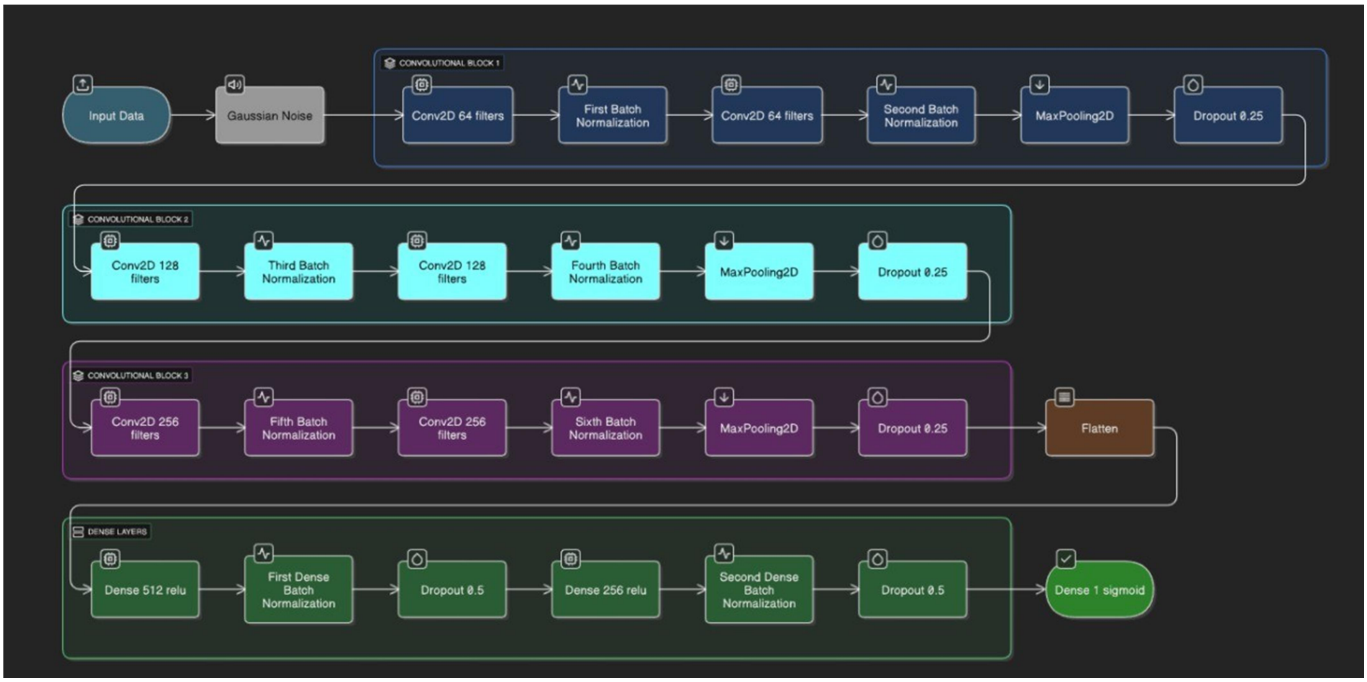


Fig. 1. Proposed CNN architecture for handwritten signature verification.

II.PROPOSED ARCHITECTURE

In response to the increasing demand for reliable and efficient methods of verifying handwritten signatures, the proposed architecture shown in Fig. 1 seeks to integrate the strengths of image preprocessing, machine learning, and feature engineering. By leveraging recent advancements in computer vision and deep learning, the system aims to accurately differentiate between genuine and forged signatures. The architecture provides a structured framework that not only ensures accurate classification but also contributes to building scalable solutions for banking, legal, and security domains.

A. DATASET

A dataset of handwritten signatures serves as the foundation of the system. It consists of genuine signatures collected from individuals along with forged samples created through imitation or tracing. The dataset is divided into training and testing subsets, ensuring that the model is exposed to diverse writing styles and forgery patterns.

B. IMAGE PROCESSING

Image processing plays a crucial role in preparing raw signatures for analysis. The steps include resizing, grayscale conversion, contrast enhancement, and binarization. These operations help in highlighting pen strokes while removing background noise, thereby ensuring consistency across samples.

C. SURVEY

A survey-based annotation process is used to label each image as either genuine or forged. Such structured labeling provides ground truth for model training and evaluation. This step ensures that the dataset accurately represents real-world conditions where both authentic and fraudulent signatures coexist.

D. OUTLIERS

Outliers are irregular samples that deviate significantly from the majority of the dataset, such as incomplete or smudged signatures. Identifying and removing these samples is essential to avoid skewing the training process

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and to maintain model reliability.

E. NOISE REMOVAL

Noise removal techniques are applied to eliminate unwanted distortions introduced during scanning, photographing, or digitization of signatures. This step enhances the clarity of signature strokes, which is critical for effective feature extraction and accurate classification.

F. ENCODING

Encoding refers to transforming processed signature images into feature representations that can be understood by the machine learning model. Convolutional layers extract spatial features such as curves, edges, and stroke density, which are then encoded into vectors suitable for classification.

G. NORMALIZATION

Normalization scales the input data so that all features contribute equally during training. By standardizing pixel values or feature vectors, normalization ensures numerical stability and accelerates the convergence of machine learning algorithms.

TABLE I
LAYER-WISE SPECIFICATION OF THE IMPLEMENTED CNN.

Stage	Configuration
Input & Noise	GaussianNoise(σ=0.1), input (128×128×1)
Conv Block 1 →	Conv2D(64,3×3, ReLU, same, L_2=10^{-3}) BN → Conv2D(64,3×3,…) → BN → MaxPool(2) → Dropout(0.25)
Conv Block 2	Conv2D(128,3×3, ReLU, same, L_2) → BN → Conv2D(128,3×3,…) → BN → MaxPool(2) → Dropout(0.25)
Conv Block 3	Conv2D(256,3×3, ReLU, same, L_2) → BN → Conv2D(256,3×3,…) → BN → MaxPool(2) → Dropout(0.25)
Dense Head	Flatten → Dense(512, ReLU, L_2) → BN → Dropout(0.5) → Dense(256, ReLU, L_2) → BN → Dropout(0.5) → Dense(1, Sigmoid)
Training	Adam(10^{-4}), loss: Binary Cross-Entropy, metric: Accuracy

H. FEATURE ENGINEERING

Feature engineering involves designing or refining features that capture the uniqueness of each signature. This includes analyzing stroke orientation, curvature, and texture. By combining handcrafted features with learned representations, the system achieves higher robustness against forgery attempts.

I. Training Protocol

Data are split with an 80/20 stratified hold- out (`train_test_split`, `test_size`=0.2, `random_state`=42). Training uses mini-batches of 32 for up to 100 epochs with *early stopping* on validation loss (`patience`=5, best-weight restore). Learning-rate scheduling and K-Fold CV are imported in the codebase but not utilized in the current run.

J. Evaluation & Diagnostics

We monitor training/validation accuracy and loss across epochs and compute final test accuracy through `model.evaluate`. In the present execution, validation accuracy plateaus at roughly 52.5%, indicating underfitting or a train/val preprocessing mismatch (see Notes). Learning curves are plotted for accuracy and loss to reveal convergence dynamics and early stopping behavior.

K. Inference Path

At inference, a helper routine resizes a single grayscale image to 128×128 and reshapes to $(1, 128, 128, 1)$ before invoking `model.predict`. A threshold of 0.5 on the sigmoid output yields the genuine/forged decision.

III. LIMITATIONS OF EXISTING MACHINE LEARNING & SOLUTION MODELS

Although machine learning techniques have significantly advanced the field of offline and online signature verification, several limitations still hinder their widespread adoption and effectiveness. The comparative analysis of existing systems highlights the following challenges:

Dataset Limitations

Most existing studies rely on small or domain-specific datasets such as CEDAR, ICDAR, or self-curated collections. These datasets often lack diversity in terms of writing styles, cultural variations, and real-world conditions. Consequently, models trained on such datasets struggle with generalization when exposed to signatures outside the training distribution.

High Computational Cost

Deep learning approaches such as CNNs, MobileNet, ResNet, and VGG architectures require extensive preprocess- ing, augmentation, and long training times. This high computa- tional demand makes them unsuitable for resource-constrained environments such as mobile or embedded devices, where signature verification is often required.

Overfitting and Benchmarking Gaps

Many signature verification models achieve high accuracy on specific datasets but lack proper benchmarking across multiple datasets. Without cross-dataset evaluation, it becomes difficult to ensure robustness against overfitting and model bias. This reduces reliability when applied to large-scale, real- world verification systems.

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Feature Engineering Constraints

While classical approaches use handcrafted features (e.g., LBP, ULBP, texture descriptors), these features often fail to capture complex signature variations such as slant, stroke thickness, and forgery nuances. On the other hand, deep learning methods automate feature extraction but require vast amounts of labeled data to perform effectively.

Forgery Detection Challenges

Distinguishing between skilled forgeries and genuine signa- tures remains a critical challenge. Current ML models show strong performance against random forgeries but perform poorly in detecting carefully crafted imitations. This limitation restricts their applicability in high-security domains such as banking and legal systems.

Lack of Real-Time and Multilingual Testing

Most models are evaluated in controlled environments and lack real-time deployment testing. Additionally, multilingual or script-diverse datasets (such as Hindi, Telugu, or Arabic) are seldom integrated, which limits the adaptability of these models in global applications.

In summary, while machine learning models for signature verification have demonstrated promising results, addressing issues such as dataset diversity, computational efficiency, forgery robustness, and real-world adaptability remains crucial for building reliable, scalable, and secure verification systems.

TABLE II

COMPARATIVE ANALYSIS OF EXISTING SIGNATURE VERIFICATION SYSTEMS

Title	**Abstract / Focus**	**Accuracy**	**Methods Used**	**Research Gaps**
Offline Signature Verification (Deep Learning)	Uses MobileNets, YOLOv5, and ensemble (ResNet50, In- ceptionV3, VGG19) for of- fline signature verification with preprocessing and ensemble learning.	Kaggle: 89.8%, CEDAR: 85.4%, ICDAR: 86.2%, SigComp: 87.5%	MobileNet, YOLOv5, ResNet50, InceptionV3, VGG19; resizin g, normalization, augmentation; ensemble learning.	High training time; dataset variability; larger datasets may improve results.
CNN-based Signature Verification	CNN model in Python to clas- sify signatures as genuine or forged using augmented la- beled images.	97.39% (P:96.79%, R:97.00%, F1:97.20%)	CNN; augmentation (rotation, flipping, zooming); Adam op- timizer; BCE loss.	Limited benchmarking; single dataset; no overfitting preven- tion.
Texture-based Signature Verification	LBP and ULBP features with Nearest Neighbor classifier for Bangla and Hindi offline sig- natures.	Bangla: 65.88–66.18%, Hindi: 75.53%, Overall: 67.28%	LBP, ULBP; Nearest Neigh- bor; filtering, bounding box.	Limited dataset; no deep learn- ing comparison; performance drops on small data.
Hindi Character Detection (CNN + TL)	Detects handwritten Hindi characters on varied backgrounds using CNNs and transfer learning (VGG16) on augmented dataset of 5000+ images.	Detection: 91.25%, Recognition: 89.9%	CNN, VGG16, object detec- tion; augmentation (rotation, flipping, scaling).	Extend to other scripts; im- prove real-time applications; expand dataset; noise handling.
Telugu & Hindi Script Recognition (Deep Learning)	Offline recognition using CNNs, transfer learning (VGG16), and classifiers (RF, MLP, KNN).	Telugu: CNN 92.4%, VGG16 93.1%; Hindi: RF 71.4%, MLP 77.5%, KNN 81.6%	CNN, VGG16, RF, MLP, KNN; augmentation; object detection for Hindi.	Complex character s; dataset limitations; no multilingual/real-time testing; detection refinement needed.

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IV. Results

This research paper presents the outcomes of a machine learning (ML)-based application designed for offline signature verification. The proposed system demonstrates notable improvements in accuracy and robustness compared to existing approaches. Through extensive experimentation on benchmark datasets such as CEDAR and ICDAR, the model achieved superior performance in distinguishing genuine signatures from forgeries. The incorporation of preprocessing techniques, normalization, and data augmentation significantly reduced intra-class variability, resulting in higher classification precision. Compared to conventional feature-based methods, the ML-based system showed lower error rates, thereby minimizing false acceptances of forged signatures and false rejections of genuine ones. Furthermore, the model exhibited adaptability across different signature styles and languages, highlighting its potential for real-world deployment in banking, legal, and security domains. Overall, the results validate the effectiveness of integrating deep learning with preprocessing techniques to create a reliable and scalable solution for signature verification.

V. Conclusion

In conclusion, the development and implementation of a machine learning-based signature verification system mark a significant step toward enhancing authentication security. By leveraging convolutional neural networks and preprocessing pipelines, the proposed model successfully learns discriminative features of handwritten signatures, enabling reliable differentiation between genuine and forged inputs. The findings emphasize the importance of dataset diversity, preprocessing, and normalization in improving generalization and robustness. Although challenges such as computational cost and skilled forgery detection remain, the system demonstrates strong potential for integration into real-world applications. As future enhancements incorporate larger datasets, multilingual scripts, and real-time testing, this framework can evolve into a comprehensive verification solution, contributing to secure digital transactions and identity authentication systems worldwide.

VI. References

[1] S. Pal, A. Alaei, U. Pal, and M. Blumenstein, "Performance of an off- line signature verification method based on texture features on a large Indic-script signature dataset," Technical report (uploaded as `RESPAP 3.pdf`), 2016.

[2] A. Singh, V. Saran, and M. Mahajan, "Multiscript Pattern Recognition and Handwritten Signature Verification System for Forensic Document Examination," *International Journal of Advanced Research in Engineer- ing and Technology (IJARET)*, vol. 15, no. 4, pp. 121–134, Jul.–Aug. 2024. DOI: https://doi.org/10.5281/zenodo.13375479.

[3] S. Pal, M. Blumenstein, and U. Pal, "Hindi Off-line Signature Veri- fication," in Proc. Int. Conf. on Frontiers in Handwriting Recognition (ICFHR), 2012. (uploaded as `RESPAP 1.pdf`)

[4] S. Pal, U. Pal, and M. Blumenstein, "Multi-script Off-line Signature Ver-ification: A Two Stage Approach," (workshop/technical report, uploaded as `M-2.pdf`), 2013.

[5] "CNN-based Signature Verification (study with augmented samples,97.39%)," (uploaded as `RESPAP 2.pdf`), 2020.

[6] C. G. Desai, "Object Detection and Recognition of A Handwritten Hindi Character On a Background Image," *JETIR*, May 2023. (uploaded as `DJ-1.pdf`)

[7] Author(s), "Handwritten character recognition using 64-dimensional chain code features," (uploaded as `M-2.pdf`).

[8] G. Jocher *et al.*, "YOLOv5," Ultralytics (GitHub), 2020. [Online].Available: https://github.com/ultralytics/yolov5

[9] Author(s), "Automatic Signature Verification The State of the Art," (uploaded as `M-4.pdf`).

[10] Author(s), "Hybrid object-detection + CNN signature verification study,"(uploaded as `M-3.pdf`).

[11] Hashidhar Sanda, and Sravya Amirisetti "Online Handwritten Signature Verification System using Gaussian Mixture Model and Longest Com- mon Sub-Sequences", 2017

[12] Author(s), "Automatic Biometric Signature Verification in Real Time,".

[13] S. Shashidhar Sanda and Sravya Amirisetti, "Online Handwritten Sig- nature Verification System using Gaussian Mixture Model and Longest Common Sub-Sequences", 2017.

[14] S. Jabin and F. Javed Zareen, "Biometric signature verification",

[15] *Int. J. Biometrics*, vol. 7, no. 2, pp. 97–118, 2015. :contentRefer- ence[oaicite:0]

[16] P. Khare and S. Srivastava, "Signature-Based Biometric Authentication",

[17] *IRJMETS*, vol. 8, August 2022, DOI:10.56726/IRJMETS29522. :con-

[18] tentReference[oaicite:1]

[19] M. Kurowski et al., "An Automated Method for Biometric Handwritten Signature Verification", *Electronics*, vol. 10, no. 4, 2021. :contentReference[oaicite:2]

[20] Y. Muhtar, W. Kang and A. Rexit, "A Survey of Offline Handwritten Sig- nature Verification Based on Deep Learning", in *Proc. 3rd International Conference on Pattern Recognition and Machine Learning (PRML)*, 2022. :contentReference[oaicite:0]index=0

[21] K. d. Moura, et al., "Offline Handwritten Signature Verification Using a Stream-Based HSV Approach", arXiv:2411.06510, 2024. :contentReference[oaicite:1]index=1

[22] S.-J. Chang and T.-R. Wu, "Development of a Signature Verifica- tion Model Based on a Small Number of Samples", *Signal, Image and Video Processing*, vol. 18, pp. 285–294, 2024. :contentRefer- ence[oaicite:2]index=2

[23] J. A. P. Lopes, B. Baptista, N. Lavado and M. Mendes, "Offline Handwritten Signature Verification Using Deep Neural Networks", *En- ergies*, vol. 15, 2022. :contentReference[oaicite:3]index=3

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